Muawiyah

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Experience

Student Placement Coordinator **Aug 2025 – Present**
Gulzar Group of Institution

- Coordinated campus recruitment drives and facilitated communication between recruiters and students.
- Improved student participation in placement activities through effective coordination and communication.

Projects

Multi Skin Diseases Classifier

Python · TensorFlow · Keras · EfficientNetV2B0 · MobileNetV2 · OpenCV · Grad-CAM · Streamlit

- Developed a multi-class skin disease classification system using **EfficientNetV2B0 transfer learning** across **9 diagnostic classes**, achieving **90% + test accuracy** and **F1-score**.
- Engineered dual-gate guardrails with a **MobileNetV2 Out-of-Domain validator**, uncertainty rejection and **Grad-CAM** visual explanations, integrated into a Streamlit dashboard.

ResearchMind — Multi-Agent Research System

Python · LangChain · Groq · Tavily · Streamlit · BeautifulSoup

- Developed an AI-powered **multi-agent research system** with specialized **Search, Reader, Writer, and Critic agents** to automate web research, content extraction, report generation, and quality evaluation.
- Integrated **Tavily web search**, **Groq LLMs**, and LangChain tool-calling with a custom **Streamlit dashboard**, enabling real-time research and downloadable structured reports.

Heart Disease Prediction

Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · KNN · Streamlit

- Built an end-to-end heart disease prediction system using **multiple Machine Learning models**, EDA, data preprocessing, and feature scaling.
- Compared model performance , achieving **85.33% Accuracy** and **87.08% F1-score**, developed a KNN-based Streamlit web application for real-time predictions.

Student Placement Prediction

Python · Pandas · NumPy · Scikit-learn · Logistic Regression · Streamlit

- Developed a placement prediction system by preprocessing academic and skill-related data and comparing multiple Machine Learning algorithms achieving **80.87% test accuracy** and **F1-score**.
- Selected **Logistic Regression** based on model performance and deployed the trained model through an interactive Streamlit web application.

Technical Skills

Programming: Python, SQL

Data Science & ML: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, EDA, Feature Engineering, Model Evaluation

Deep Learning & AI: TensorFlow, Keras, OpenCV, CNN, RNN, LSTM, Transfer Learning, NLP

Generative AI: LangChain, Hugging Face, LLMs, RAG, Embeddings, Vector Databases, Prompt Engineering

Development & Deployment: FastAPI, Streamlit, Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Education

B.Tech in Data Science **2023 – Present**
IK Gujral Punjab Technical University, Punjab CGPA: **8.0**

Senior Secondary (12th) — SAS Senior Secondary School 2023

Secondary School (10th) — RSAS High School 2021

Certifications

- **Prompt Engineering (LinkedIn)** – Learned techniques to design clear and effective prompts for AI/LLMs.
- **Introduction to Data Science (LinkedIn)** – Covered fundamentals of data analysis, visualization, and machine learning.
- **Deloitte Data Analytics Job Simulation (Forage)** – Completed hands-on data analysis and forensic technology tasks.